DIDACTIC SYNTHESIS · PROBING RESULTS & CROSS-REFERENCE

Nasal Place Assimilation in wav2vec2

The probing results on English nasals, integrated and extended with a nine-step articulatory continuum between [n] and [m] in *clean pan*, probed layer by layer in two wav2vec2 encoders—and crossed, dimension by dimension, with the perception experiments of Pouw et al. (2024).

Surface Dominance · Letter Supervision · The Two-Stage Account

Probing: wav2vec2-base · wav2vec2-base-960h (CTC-tuned) · 13 layers, linear probes
Synthesis: covtl-pipeline v1.1.0 · VocalTractLab · speaker JD3 · λ-morph continuum, full CC selector (kit v3.1)

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Abstract

This document synthesizes, in a didactic form, the experimental results of our research program on English nasal place assimilation in wav2vec2, and crosses them systematically with the published findings of Pouw, de Heer Kloots, Alishahi, and Zuidema (2024) on the perception of phonological assimilation by neural speech recognition models. Our program synthesizes canonical and assimilated pronunciations of the same utterances with an articulatory engine, so that the two variants of a pair differ in exactly one consonantal gesture and are aligned frame by frame; linear probes then read the phone identity out of every layer of two encoders, the CTC-tuned wav2vec2-base-960h and the purely self-supervised wav2vec2-base. The integrated results are: surface dominance (assimilated segments are encoded as the phone they have become in roughly 80 percent of token-layer decisions); a partial drift toward the underlying phone in the deepest layers of the CTC-tuned model only, which we attribute to letter supervision; no phonological licensing in the unlicensed controls; and a context effect showing that mid-depth representations are conditioned by surrounding segments. A new experiment, reported here for the first time in this edition, rotates the polar *target* of the C1 nasal of tautosyllabic clusters toward the C2 place in nine steps, on six frames—the reference model of the `covtl-lambda` repository (v1.2.0) extended so that under λ the whole CC selector is taken into account, as in the base model (four run-time guarantees hold C2, the context, the velum and the glottis bitwise canonical, and the lips join the C1 gesture with weight λ)—and probes every intermediate token: the phonetic layers hold the underlying /n/ at every degree of every frame (the category boundary of the earlier motor-blend editions no longer exists, and it does not return even when the lips close), what they track instead is the coarticulation of the canonical cluster and its cleanup under the velar rotation, and the licensed context exerts a unidirectional, supervision-dependent labial reading in the deepest CTC layers—a nasal before a labial reads toward [m] regardless of the gesture, strongest on a frame that contains no /m/ segment at all, and unchanged when the assimilated frame's own lips actually close. The cross-reference shows convergent, complementary, and occasionally divergent answers from the two paradigms, and motivates a two-stage account of assimilation processing in wav2vec2.

1 Introduction: Two Programs, One Question

When a speaker of English says *clean pan*, the word-final /n/ of *clean* very often takes the place of the following /p/, and the phrase surfaces as "clea[m] pan". A listener who recovers the intended word *clean* from this signal is said to *compensate* for place assimilation. The phenomenon is a classical test case for the relation between what a speaker intends (the underlying form, /n/) and what the waveform carries (the surface form, [m]), and it has been studied in humans for three decades with ever more refined behavioral paradigms [4, 7, 9, 10]. The question this document pursues is what a modern neural speech encoder does with the same material: when the acoustics say [m] but the lexicon and the phonology say /n/, which of the two does the model's internal representation keep?

Two research programs have recently addressed this question, and they do so from directions that are strikingly complementary. The first is external: Pouw and colleagues [17] fed natural, psycholinguistically validated stimuli to the CTC-tuned wav2vec2 encoder and asked how assimilation shapes its transcriptions, then localized and causally intervened on the model components that carry compensation. The second is ours: we synthesize the stimuli ourselves, with an articulatory engine that replaces exactly one consonantal target while leaving every other gesture—and the entire frame grid—untouched, and we then probe every layer of two wav2vec2 encoders with linear classifiers [2, 3]. Their material is natural but only statistically controlled; ours is synthetic but causally controlled. They measure behavior (output transcriptions) and then open the box; we measure representations directly at fixed articulatory ground truth. The two programs even share their favorite sentence, *clean pan*, which makes the confrontation unusually direct.

This document has three goals, and it is organized to serve them in order. First, it *integrates* the results of our previous report on nasals (the probing stage of the program) into a self-contained narrative, so that a reader who has not seen that report can follow everything (Sections 2–5). Second, it *extends* those

results with a new experiment that probes a nine-step articulatory continuum between [n] and [m], in a licensed and an unlicensed context, on both encoders (Section 6). Third, it *crosses* our findings with those of Pouw et al., dimension by dimension, distinguishing convergence from complementarity and from genuine divergence (Section 7). The synthesis supports a two-stage account of assimilation processing in wav2vec2 (Section 8), which we state in didactic form and confront with the human literature.

A note on style: this is a didactic text. Every technical notion is defined before it is used, every figure is read panel by panel, and every claimed number is traceable to an experiment. Readers fluent in speech-representation analysis can skim Sections 2–3 and start in earnest at Section 4; readers new to the domain should read in order. SAMPA transcriptions are set in typewriter face (`klin` for the canonical *clean pan*, `klim` for the assimilated variant), IPA where the argument is phonetic.

2 Background: Assimilation, Viability, Compensation

2.1 Underlying and surface forms

Phonology distinguishes the *underlying form* of a morpheme—the mental representation that supports lexical contrasts—from the *surface form* that is actually pronounced in a given context. English place assimilation is the canonical example of a context-sensitive rewrite: a word-final coronal consonant (/n/, /t/, /d/) takes the place of articulation of a following non-coronal consonant, so underlying /n/ surfaces as [m] before labials and as [ŋ] before velars [12]. The process is productive, largely exceptionless in fast speech, and—crucially for everything that follows—*recoverable*: listeners are normally unaware that anything happened at all. That invisibility is the puzzle. If the surface signal contains [m], what licenses the listener to hear /n/? And if a machine learns from the signal alone, or from the signal plus spellings, what does it internalize about the pair?

Two idealized answers structure the field. A purely *lexical* mechanism treats assimilation as noise: it matches the incoming [m] against stored word forms and lets the lexicon decide (“cleam” is not a word, “clean” is). A purely *phonological* mechanism treats assimilation as a rule: it checks whether the phonological context licenses the change and undoes it if so. The two mechanisms make different predictions about *unlicensed* assimilations—surface forms in which the place change has occurred but the context does not trigger it, such as “clea[m] tan” before the alveolar /t/. A lexical mechanism is indifferent to the context (“cleam” is a non-word either way); a phonological mechanism restores /n/ only when the context licenses the change, so it should compensate less in unlicensed than in licensed contexts. Human listeners behave phonologically in this sense [4, 9], with documented lexical modulations [10].

2.2 Complete, incomplete, and gradient assimilation

Real assimilation is not always total. Electropalatographic and articulometric studies show that word-final coronals before velars and labials range from fully assimilated (no trace of the coronal gesture) through partially assimilated (overlapping gestures, reduced but present coronal contact) to essentially canonical [5, 6, 15]. Incomplete assimilation leaves acoustic traces of the underlying place, and listeners use them: compensation is facilitated when residual coronal cues are present [7]. This gradience matters for two reasons. First, it means that the underlying/surface opposition is a continuum at the articulatory level, not a binary. Second, it motivates the experiment of Section 6: if one can synthesize controlled intermediate articulations, one can ask where along the continuum a given representational system locates its category boundary, and whether anything phonological (the following consonant) moves that boundary. Pouw et al. deliberately restricted their stimuli to complete assimilation, precisely so that their model could not rely on residual acoustic cues [17]; our continuum addresses the complementary regime.

2.3 Compensation in machines

Before wav2vec2, the machine question was asked of classical systems: **(author?)** [13] showed that hidden-Markov-model recognizers with context-sensitive acoustic models and phonotactic language models reproduce human compensation patterns for viable and unviable assimilated forms, without a lexicon.

End-to-end neural models were unexplored until Pouw et al., whose study we now summarize in detail, because our cross-reference rests on it.

3 The External Anchor: Pouw et al. (2024)

The study we cross-reference is "Perception of Phonological Assimilation by Neural Speech Recognition Models" by Charlotte Pouw, Marianne de Heer Kloots, Afra Alishahi, and Willem Zuidema, published in *Computational Linguistics* **50**(4):1557–1585 [17]. This section is a didactic walkthrough of its design and results; all numbers quoted here come from that article. The study asks one question—does the neural ASR model wav2vec2 compensate for place assimilation the way humans do?—and answers it with four instruments: three computational behavioral experiments, a probing analysis, and a set of causal interventions. The model at the center is the same CTC-tuned encoder our program probes, `facebook/wav2vec2-base-960h`: seven convolutional layers followed by twelve transformer layers, pretrained on 960 hours of unlabeled LibriSpeech, then fine-tuned to map frame-level representations to written characters with a connectionist temporal classification objective [1, 11]. The choice matters: CTC fine-tuning teaches the model that the *text* of an assimilated token is the spelling of its underlying form, so any compensation the model displays could come from phonology, from the lexicon, or from letter statistics.

3.1 Three behavioral experiments

Experiment 1 borrows the stimuli of **(author?)** [4]: 16 target words covering six assimilation types (/n/[m], /d/[b], /t/[p] before labials; /n/[ŋ], /d/[g], /t/[k] before velars), each embedded in three contexts—viable (*clea[m] pan*), unviable (*clea[m] spoon*), and control (*clea[n] fork*)—and three carrier sentences, for 144 stimuli in all, recorded by a speaker instructed to produce complete assimilation. The measured quantity is the *compensation rate*: the percentage of stimuli whose transcription contains the underlying consonant. The result is double-edged. The model compensates markedly more often in viable than in unviable contexts, which is the signature of sensitivity to phonological context. But the unviable rate is still about 40 percent, and the model compensates more often than the human participants of the original study did. Two follow-up observations qualify the picture. First, compensation correlates with how often the target-plus-context bigram occurred in the fine-tuning text (Spearman $\rho(76) = .40$, $p < .001$), although bigram frequency cannot be the whole story: the model compensates *mad brother* (a bigram never seen in training) and refuses to compensate *great match* (seen 43 times). Second, the model sometimes "compensates" into a different word: it transcribes *we[p] pants* as *wept pants* and *plai[ŋ] condoes* as *playing condoes*, hinting at lexical knowledge, at knowledge of other reduction processes, or both.

Experiment 2 removes the lexical escape route. Using the stimuli of **(author?)** [10], the assimilated surface form is itself a real word: *ru[m]* can be the ordinary word *rum* or the assimilated form of *run*, and the following context decides whether the assimilation is viable (*ru[m] picks*) or unviable (*ru[m] does*). With the non-word override unavailable, compensation rates drop sharply relative to Experiment 1, though the viable–unviable difference persists. In plain terms: when the waveform spells a real word, the model mostly transcribes that word, and the phonological context only tips the balance.

Experiment 3 adds semantics. The *run/rum* sentences are preceded either by silence, by a biasing context sentence ("It's best to start the day with a burst of activity."), or by an unrelated sentence. Human listeners use such biasing contexts to keep both lexical candidates alive [10]. Wav2vec2 does not: the compensation rate is statistically indistinguishable across the three preceding contexts, and the frame-level probability of the underlying consonant is the same under biasing ($M = 0.14$) and neutral ($M = 0.17$) context. The authors attribute this insensitivity to the model's frame-level objectives; a preliminary comparison with the encoder–decoder model Whisper [18], whose decoder predicts text autoregressively, compensates more often overall but is still indifferent to preceding sentential context.

3.2 Probing: where compensation lives

The behavioral experiments tell us *that* the model compensates, but not where in the network the underlying interpretation forms. For that, Pouw et al. extract the frame representation at the position of the assimilated consonant (located via the model's own transcription) and hand it to a binary linear probe—logistic regression trained on TIMIT frames [8]—that must distinguish the underlying from the surface phoneme of the assimilation type at hand. The probes are accurate (0.75–1.00, best in middle-to-final layers), and their layer-wise probabilities tell a clean story. When the model's transcription *did* compensate, the probe sees the surface phone in early-to-middle layers and a growing preference for the underlying phone in the final layers: somewhere between the middle and the top of the stack, the representation migrates from what was said to what was meant. When the transcription did *not* compensate, the probe sees the surface phone at every layer. Controls (unassimilated inputs) read as the underlying phone throughout. In a footnote of considerable importance for our cross-reference, the authors repeated the analysis with the purely self-supervised `facebook/wav2vec2-base` and observed the same early pattern but a *converging* rather than diverging final-layer profile across conditions—the untuned model does not sharpen the underlying interpretation the way the CTC model does [17], consistent with the different training objectives [16].

3.3 Causal interventions: how context reaches the consonant

The last instrument is the sharpest. Taking a viable/unviable pair with known opposite outcomes (*thi[m] packet* transcribed as *thin packet*; *thi[m] leaflet* transcribed as *thim leaflet*), the authors run both stimuli through the model, then re-run the unviable one while overwriting the output of a chosen component—an attention head or an MLP block in a given layer, at a chosen time position—with the output that the same component produced during the viable run. If the overwrite flips the predicted character from <*m*> to <*n*>, that component carries the context cue that compensation needs. The result is a two-act mechanism. Early MLPs (layers 1–5) write the decisive cue into the residual stream, but only when overwritten at the position of the *context word*—and the effective region is exactly the critical context phone (*[l]* or *[p]*), not the single critical frame and not the whole word. Later MLPs (layers 6–12) can still flip the decision, but only when overwritten inside the assimilated word itself: by then the cue has already been moved to the target position. Two attention heads (head 3 of layer 4 and head 2 of layer 5) do the moving, gathering information from the critical context phone and writing it onto the assimilated frames. A single phone's worth of context, propagated by a small circuit, is thus sufficient to flip the transcription of the assimilated consonant. Table 1 condenses the four instruments and their findings for later reference.

Table 1: The four instruments of Pouw et al. (2024) in brief.

Instrument	Contrast	Headline finding
Exp. 1 (ASR output)	viable vs. unviable vs. control, non-word assimilated forms	compensation higher in viable contexts; still ≈40% in unviable; bigram effect $\rho = .40$; lexical over-rides (*wept*, *playing*)
Exp. 2 (ASR output)	real-word surface forms (*rum*/*run*)	much lower compensation overall; viable–unviable difference persists
Exp. 3 (ASR output)	biasing vs. neutral vs. random preceding sentence	no semantic effect on compensation ($M = 0.14$ vs. 0.17), unlike humans
Probing + interventions	layer-wise underlying/surface preference; interchange of head/MLP outputs	surface-to-underlying shift in final layers when the output compensated; pretrained model converges instead; early MLPs at the context phone + heads L4H3/L5H2 carry the cue

4 Our Experimental Program

4.1 Articulatory synthesis as a causal instrument

Our program makes the opposite trade-off from Pouw et al.: instead of natural speech with statistical control, we use synthetic speech with causal control. The covtl-pipeline engine [2] drives VocalTractLab [3] through a polar control plane: each phoneme is a target point $z = \rho e^{i\theta}$ in a two-dimensional articulatory summary space that projects onto the 19-parameter vocal-tract state, and phoneme identity is therefore a single point edit. Replacing the target `n` of `klin.pan` ("clean pan") by `m` regenerates the consonantal branch toward a labial closure while the vocalic branch, the phoneme boundaries, and the frame count are held identical; the engine's coarticulation logic does the rest, and the resulting pair is aligned frame by frame (624 tract frames at 400 Hz, 1.56 s of audio at 44.1 kHz, identical nasal windows 0.58–0.68 s). In the causal vocabulary of intervention analysis, the pair is a manipulation of exactly one factor—the closure gesture of the target consonant—with everything else held fixed. We verified in the previous stage that the vocalic branches are bitwise identical between variants and that the tract-parameter difference is confined to the consonant and its transitions. One engine limitation needed a documented patch: the velar nasal `N` had no closure gesture of its own, so a velar closure (dorsum to soft palate, velum open) was added as a three-line selector patch; with it, the assimilated [ŋ] of `kliN.kan` is articulatorily identical to the canonical [ŋ] of the training word `kiN` (tongue-body and tongue-tip positions equal to within 10^{-6} of a parameter unit).

4.2 The 52-item base

The probing base comprises 52 two-word sentences in four groups (Table 2). Six *viable pairs* contrast a canonical with an assimilated variant: *clean pan* and *thin packet* (/n/[m] before /p/), *mean boat* (/n/[m] before /b/), *clean can* (/n/[ŋ] before /k/), *that pen* (/t/[p] before /p/), and *that cake* (/t/[k] before /k/). Two *unlicensed controls* apply the same substitutions before an alveolar that does not trigger assimilation: *clean tan* (`klim.tan`) and *that time* (`Dap.tajm`); these are our exact counterparts of Pouw et al.'s unviable condition, and unlike theirs, ours is frame-aligned with its licensed twin down to the nasal window. Finally, 36 *training sentences* (six per class for /n/, /m/, /ŋ/, /t/, /p/, /k/) supply balanced material for the probes: monosyllables ending in the target consonant, each followed by one of the three carriers *pan*, *tan*, *can*, so that following-context is balanced across classes and no test lexical item appears in training.

Table 2: The eight test pairs of the 52-item base (canonical SAMPA, process, licensing context). All variants of a pair share their frame grid and phoneme boundaries.

Pair	Sentence	Process	Context
P1	clean pan (`klin`/`klim`)	/n/[m]	labial (/p/), licensed
P2	mean boat (`min`/`mim`)	/n/[m]	labial (/b/), licensed
P3	clean can (`klin`/`kliN`)	/n/[ŋ]	velar (/k/), licensed
P4	thin packet (`Tin`/`Tim`)	/n/[m]	labial (/p/), licensed
P5	that pen (`Dat`/`Dap`)	/t/[p]	labial (/p/), licensed
P6	that cake (`Dat`/`Dak`)	/t/[k]	velar (/k/), licensed
NV1	clean tan (`klin`/`klim`)	/n/[m]	alveolar (/t/), unlicensed
NV2	that time (`Dat`/`Dap`)	/t/[p]	alveolar (/t/), unlicensed

4.3 Encoders, probes, and read-outs

Two encoders were probed: the same `facebook/wav2vec2-base-960h` that anchors Pouw et al., and the purely self-supervised `facebook/wav2vec2-base`—the pair that their footnote 5 compares in passing and that we make a main design axis, because the difference between the two checkpoints is exactly the letter supervision. Audio is resampled to 16 kHz, normalized per utterance, and passed

through each model with all thirteen hidden states (CNN plus T1–T12) recorded. For every sentence, the frames whose centers fall in the target-consonant window are averaged, giving one 768-d vector per layer per token. Linear probes (multinomial logistic regression, one per layer per manner class: nasals /n/,/m/,/ŋ/ and stops /t/,/p/,/k/) are trained on the 36 canonical training sentences and validated leave-one-sentence-out; they then deliver a verdict (argmax class) and class probabilities for every test token. Three training-free analyses complete the read-out: cosine distances to class centroids, a frame-wise timecourse of those distances, and a *causal delta* $\delta^{(\ell)}(t) = 1 - \cos\big(h^{(\ell)}_{\text{canon}}(t), h^{(\ell)}_{\text{assim}}(t)\big)$ that measures, layer by layer and frame by frame, how much the single-gesture intervention moves the representation. The probes' cross-validated accuracy is the instrument check: nasals are decoded at 1.00 (T2–T4, CTC model) and 0.94–1.00 across low and middle layers of the base model, stops at 0.83–0.89, which licenses every claim that follows.

5 Results of the Probing Campaign

This section integrates the results of the previous report on the nasal-and-stop probing base; every number can be traced to the per-layer classification records of that stage.

5.1 Surface dominance, layer by layer

Figure 1 shows the central object: the verdict of the place probe for every assimilated and unlicensed token, layer by layer, for both encoders. The aggregate picture is unambiguous. At the CNN layer the surface rate is 6/6 for the viable assimilated tokens in both models, it stays at 4–5/6 through the middle layers, and across all 156 token-layer decisions (6 pairs $\times$ 13 layers $\times$ 2 models) roughly 80 percent read the manipulated consonant as the phone it has acoustically become. *clea[m] pan* is /m/ with confidence $p \approx 0.72$–0.83 from the CNN through T6 of the CTC model. The self-supervised model recovers toward full surface dominance at the top (5/6 at T10–T12): given the waveform alone, wav2vec2 builds representations that follow the acoustics. This is the encoder-side premise that makes the whole program interpretable, and it matches, at the representation level, the acoustic-primacy results of the human perception literature [7].

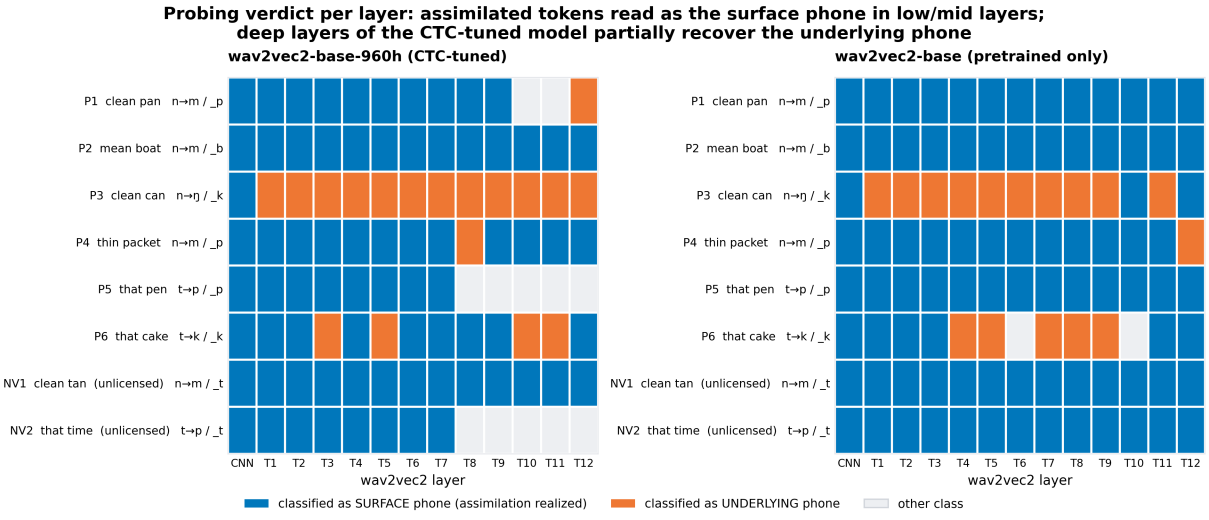


Figure 1: Probe verdict for every assimilated/unlicensed token, layer by layer, for both encoders (previous report). Blue = surface phone, orange = underlying phone, gray = third class. NV1/NV2 are the unlicensed controls.

5.2 The letter-supervision effect

The models part ways only in their deepest layers, and they do so in the direction of the CTC model's training signal. For P1, the verdict dissolves late in the fine-tuned encoder: a near tie at T8 ($p_m = 0.50$ against $p_n = 0.27$), a flip to the third class [ŋ] at T10 ($p = 0.45$), and the underlying /n/ winning at T12 with $p_n = 0.52$. The self-supervised model casts the same token as /m/ in 13 of 13 layers. P5 (*tha[p] pen*) shows a second flavor of the same drift: the long homorganic [pp] closure is re-segmented as a single strong stop whose place is reassigned to /k/ at T8–T12 (p up to 0.89). And the effect even reaches *canonical* speech: P2-canonical (*mean boat*, underlying [n] before /b/) drifts to /m/ in layers T7–T12 of the CTC model (underlying rate 0.46), exactly where the training text would have taught the model that word-final /n/ before a labial is often spelled with the labial's letter. The self-supervised model shows none of these effects. We summarize the pattern as a *letter-supervision effect*: pairing assimilated acoustics with underlying spellings during CTC fine-tuning installs a late, partial, statistics-driven pull toward the underlying category, with re-segmentation and expectation artifacts as side products.

5.3 No licensing at the verdict level

The unlicensed controls answer the phonological question directly. NV1 (*clea[m] tan*) is classified as /m/ in 13 of 13 layers in both models; NV2 (*tha[p] time*) as /p/ in 13/13 layers of the base model (8/13 in the CTC model, whose late layers drift to /k/ exactly as in P5). An [m] before /t/ is represented exactly like an [m] before /p/: neither encoder implements anything resembling an assimilation-licensing rule. Whatever "knowledge of assimilation" the CTC model displays is statistical, not rule-based—a point to which the cross-reference returns, because it is here that our verdict-level result and Pouw et al.'s output-level result are most instructively different.

5.4 The P3 exception: mid-depth representations are context-conditioned

P3 (*clea[ŋ] can*) breaks the surface-dominance pattern from the very first transformer layer: the assimilated [ŋ] is classified as /n/ from T1 on, in both encoders ($p_n \approx 0.71$ at T4), even though its articulation is identical to the canonical [ŋ] of the training words (a three-way contest at the CNN layer, [ŋ] 0.40 / [m] 0.37 / [n] 0.23, resolves toward /n/). The geometry makes the exception visible (Figure 2): in the PCA of layer-T4 window means, every other assimilated token has crossed onto the surface side of the place continuum, but P3's star lands next to the /n/ cluster. Since the segment itself cannot explain the verdict (it is bit-identical to training [ŋ]s that are read as /ŋ/), the explanation must be the context: the lateral onset of *clean* and the velar /k/ of *can*, which wav2vec2's widening receptive fields integrate into the window representation. By mid-depth, the encoder's "phone" representation is already a context-conditioned object. This is our representational counterpart of the mechanism that Pouw et al.'s causal interventions expose from the other side: their heads L4H3/L5H2 *write* context information into the assimilated frames; our P3 shows the written result.

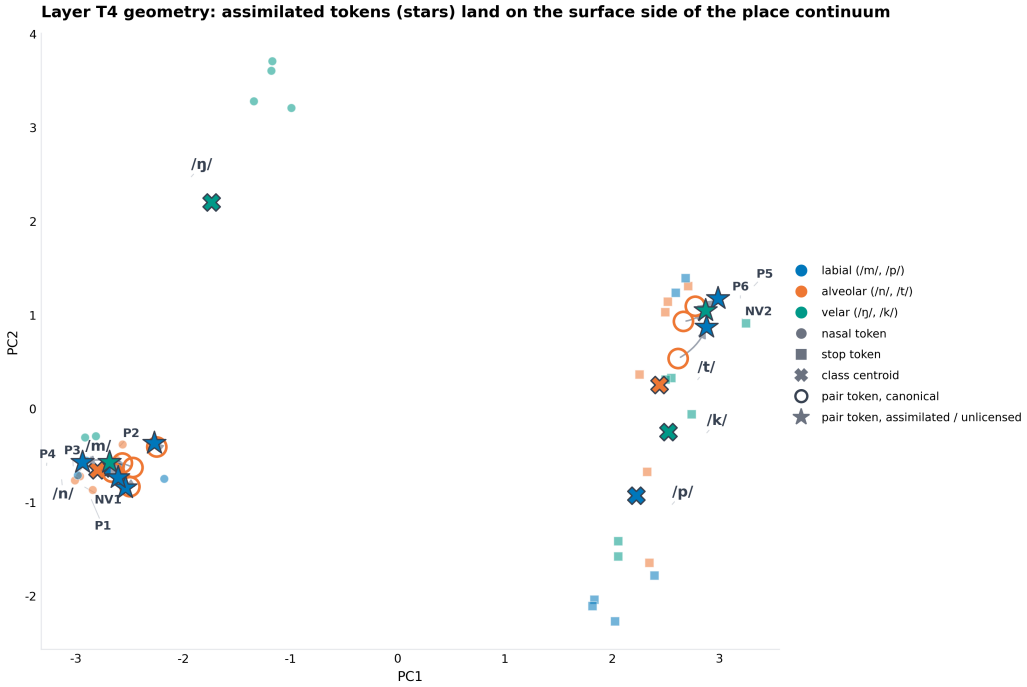


Figure 2: PCA of the layer-T4 window means (CTC model, previous report). Training tokens are colored by place, circles show pair-canonical positions, stars the assimilated/unlicensed outcomes, arrows the canonical-to-assimilated displacement. P3's star is the one that stays on the /n/ side.

6 New Experiment: The Assimilation Continuum

6.1 Design

The experiments above compare two endpoints, canonical [n] and fully assimilated [m]. Natural speech, however, fills the space between them [5, 6], and Section 2 argued that the intermediate regime is where acoustic and phonological sources of information can be dissociated. This edition of the experiment constructs that intermediate regime on the revised reference model: the λ effect of the repository `covtl-lambda` (v1.2.0) acts on the *polar target* of the C1 nasal of a tautosyllabic cluster,

$$\theta_{C1}(\lambda) \;=\; \theta_{C1} \;+\; \lambda \cdot \mathrm{wrap}\big(\theta_{C2} - \theta_{C1}\big), \qquad \lambda \in \big\{0, \tfrac{1}{8}, \tfrac{2}{8}, \ldots, 1\big\}, \tag{1}$$

and the engine's own syllabic trajectory builder is re-run per degree with the C1 node retargeted through a run-time override. The continuum is therefore not a blend of two state sequences but a family of *re-planned gestures*: at $\lambda = 0$ the synthesis is bitwise identical to the canonical cluster; at $\lambda = 1$ the C1 target has reached the C2 coordinates—the place-assimilated cluster C1'C2—while C2 itself, the glottal block and the velum never move, and the lips *join* the gesture: this edition takes the whole CC selector into account under λ, as in the base model, whose cluster machinery already drives the union of the two consonant selectors. Four run-time guarantees plus three full-selector checks are recorded per synthesis ($\lambda = 0$ bitwise canonical; θ equality at $\lambda = 1$; glottal block bitwise constant; context outside the C1-incident arcs bitwise canonical; the recruitment confined to those arcs; for the unlicensed frames, whose C2 selectors are subsets of sel(/n/), a verified bitwise no-op), and all held for all 54 syntheses. The experiment is realized for nine values of λ on six frames: the licensed pool (*clean pan*, *mean boat*, *thin packet*; n before the labials /p/, /b/, /p/, a $+140°$ target rotation) and the unlicensed pool (*clean tan*, *clean can*, *mean goat*; before /t/, /k/, /g/, $+15°$ to $+75°$)—the same licensed-versus-unlicensed contrast that separates Pouw et al.'s conditions. The two pools differ in exactly the C2 of the cluster—the same minimal contrast that separates their viable from their unviable condition. The ground truth is gestural, not acoustic (Figure 4a): the rotation is carried by the jaw and the tongue body (over P1, jaw angle $-2.34 \rightarrow -3.28°$, tongue-body x $3.96 \rightarrow 2.51$ cm, y $0.20 \rightarrow 0.66$ cm), while the velum stays *bitwise*

constant along every continuum and the lips are recruited in proportion to λ on the labial-C2 frames (lip opening at the C1 target point of P1: $0.61 \rightarrow 0.46 \rightarrow 0.16 \rightarrow 0.00$ cm at $\lambda = 0/0.5/0.75/1$, with protrusion rising to the base model's own labial value)—the full CC selector realizes the interpolated target, so the assimilated C1 is a nasal whose gesture *becomes* C2's, closure included, degree by degree. Each of the 54 stimuli was synthesized with the same engine, resampled and normalized exactly like the base material, passed through both encoders, and classified by the unchanged nasal probe instrument (leave-one-sentence-out accuracies identical to the stage-1 report), so the continuum is measured by the same probes as the endpoints.

Two facts secure the validity of the new material. First, the four run-time guarantees and three full-selector checks above are structural, checked per synthesis against the engine's own canonical output—$\lambda = 0$ is *by construction* the canonical cluster, not a re-synthesis of it, so no endpoint anchoring against the stage-1 items is needed or meaningful (the stage-1 dotted citation forms differ from the cluster by design; the machine-readable distance between the tiers, up to 0.86 in $P(\mathrm{n})$, is reported as the coarticulation signature of the tautosyllabic cluster). Second, all nine degrees of all six frames were probed in a single run with one instrument, so every cross-section comparison of this document—anchors, production degrees, dose-response—is a slice of one experiment rather than a family of re-runs, and the leave-one-out accuracies of the stage-1 report remain the honest generalization measure.

6.2 Where the category boundary went

Figure 3 maps the probe probability of the underlying /n/ over the layer $\times$ λ plane, for both encoders, the licensed frame P1 against the unlicensed velar frame NV3. Read first along the λ axis at the phone-encoding layer T4 (Figure 4b): the probability of /n/ *never falls below 0.5 anywhere on the revised continuum*. The per-frame T4 minima stay between 0.46 and 0.84, the pooled pool-level curves never cross, and the category boundary that anchored the previous edition's reading ($\lambda^* \approx 0.7$–0.9) does not exist. Three facts deserve the reader's attention. First, the boundary's disappearance is articulatory, not statistical: the motor blend of the previous editions closed the lips inside the C1 window, producing real [m] acoustics that the phone code tracked; the target-level rotation moves the jaw and the tongue, and the resulting acoustics never leave the /n/ basin. The full-selector condition of this edition closes the lips again—in proportion to λ, reaching the base model's own labial closure at $\lambda = 1$—and the boundary still does not return: the encoder is not merely tolerating an incomplete gesture; it keeps the underlying reading through a completed labial closure whose tongue keeps aiming at the rotated target. Second, what T4 tracks instead is the *coarticulation cleanup*: on the velar-C2 frames the probability of /n/ rises monotonically with λ (NV3: $0.81 \rightarrow 0.90 \rightarrow 0.95$; pooled unlicensed $\rho(\lambda, P(\mathrm{n})) = +0.90$ at T4), because the canonical cluster's murmur is labial-coarticulated—the CNN reads it as /m/-ish at $\lambda = 0$ ($P(m) = 0.54$ pooled licensed)—and rotating the C1 target away from the labial blend moves the murmur toward the trained /n/ centroid. The continuum is graded, as **(author?)** [6] describe articulatorily, but its gradient at the phonetic layers is a coarticulation gradient, not an assimilation gradient. Third, and most important for the cross-reference: the only categorical surface event of the continuum lives at the deepest CTC layers and the very end of the rotation—at $\lambda = 1$ the licensed pool's deep cells read [m] (pooled $P(\mathrm{m})$ at T10–T11 of the CTC model: 0.57–0.66) —and the full-selector condition reallocates it: the crossing is carried by P2 and P4, whose cluster context produced it already at $\lambda = 0$, while P1, whose lips now close, *loses* its endpoint crossing (T10–T11: $P(\mathrm{m}) = 0.24/0.09$ against $0.44/0.53$ with the lips blocked)—so any context sensitivity the encoders display is confined to exactly the layers where letter supervision lives and is driven by the cluster context, not by the surface closure, and never by the phone code that sets boundaries.

6.3 A labial-context reading in the deep CTC layers

The heatmaps' bottom rows are where the two columns finally separate, and only in the CTC-tuned model. Figure 4c reads this as $P(/[m]/)$ at T10 for the licensed pool against the unlicensed pool, degree by degree. In the deepest CTC layers (T10–T11) the licensed pool carries a [m] reading that is *unidirectional* in λ: it is already present at $\lambda = 0$ (pooled $P(m)$ at T10 of 0.46–0.54, driven by P2 and P4), and the full-selector

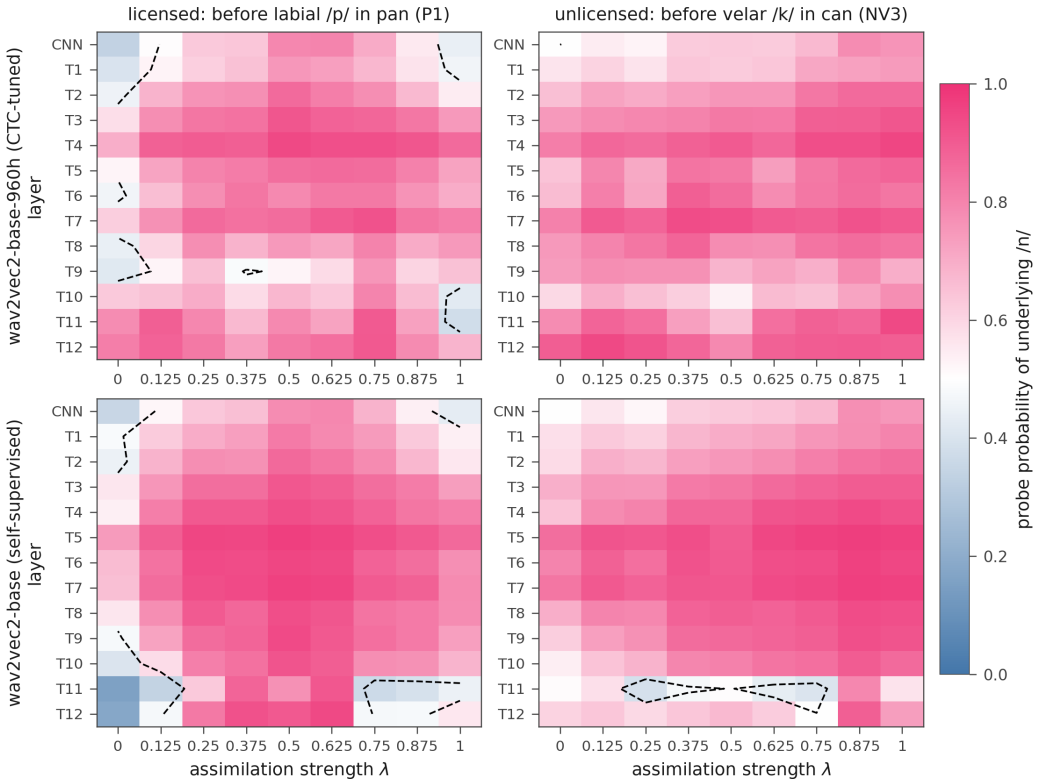


Figure 3: The assimilation continuum, probed. Color encodes the nasal probe's probability of the underlying /n/ for every layer (CNN, T1–T12) and every target-rotation strength λ (Eq. (1)), in the licensed frame P1 (*clean pan*, labial C2) and the unlicensed velar frame NV3 (*clean can*), for both encoders. The dashed black contour marks $P(/n/) = 0.5$ where it exists—at the phonetic layers it does not: the continuum has no category boundary, and in the licensed column the contour now survives only transiently (P1, around $\lambda \approx 0.4$ and at the T10–T11 endpoint of the deepest layers); the sustained endpoint crossing lives on P2/P4, outside this figure.

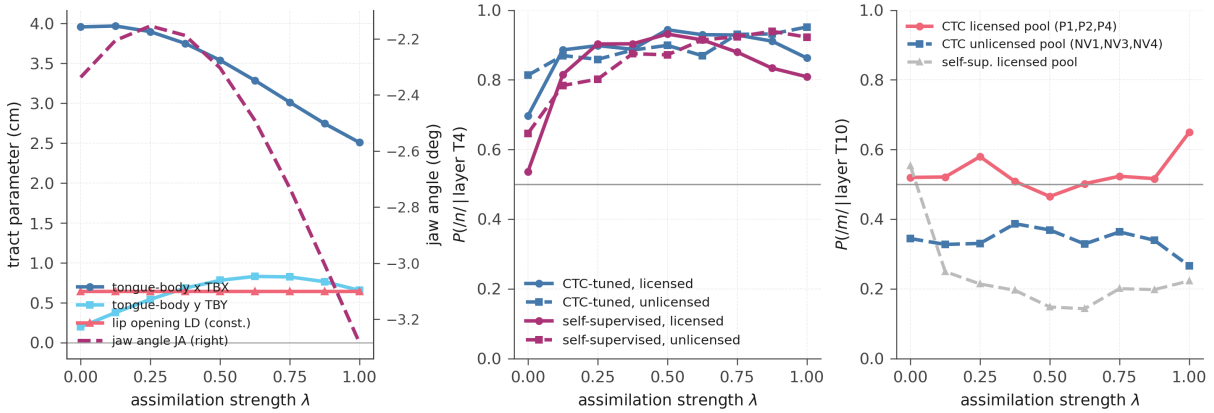


Figure 4: Reading the continuum. (a) Articulatory ground truth at the midpoint of the C1 window (P1): the gesture carriers vary with λ—tongue-body x and y drift, the jaw angle (right axis) drops—and the lip opening joins the gesture with weight λ (the full CC selector; the dotted vertical marks $\lambda = 0.5$). (b) Probability of the underlying /n/ at layer T4 as a function of λ: every curve stays high; the velar frame NV3 rises with λ (coarticulation cleanup) and no curve crosses 0.5. (c) The late-layer [m] surface: $P(/[m]/)$ at layer T10 of the CTC model, licensed pool (mean of P1, P2, P4) against the unlicensed pool—the assimilated reading is a late-layer, labial-C2 phenomenon, present already at $\lambda = 0$ on P2/P4 and indifferent to the recruited closure of P1, absent from the unlicensed pool and from the self-supervised model.

condition shows how gesture-blind it is: with the lips blocked (v3.0) the rotation sharpened P1's reading ($0.24 \rightarrow 0.44$ at T10, crossing at T11); with the lips *recruited*—a true labial closure through most of the C1 window—P1's reading reverts to its baseline ($0.24 \rightarrow 0.24$ at T10, $P(\text{m}) \leq 0.09$ at T11) while P2/P4 keep theirs. The model has learned that a nasal in a tautosyllabic cluster before a labial is where /n/ is often spelled with the labial's letter, and its deepest layers read that context as [m] *regardless of the gesture—and regardless of the closure*: at $\lambda = 0$ the C1 is a canonical alveolar closure, at $\lambda = 1$ its lips are closed on the labial place, and the reading is the same wherever the context already produced it. This is the continuum generalization of the P2-canonical attraction reported in the probing campaign, and it is *not* a compensation effect: it suppresses the underlying interpretation in exactly the context where compensation would enhance it. The decisive frame is P4 (*thin packet*): it contains no /m/ segment anywhere, yet its T10 reading is [m] along the whole continuum ($P(m)$ up to 0.82 at $\lambda = 1$)—the reading is carried by the cluster context, not by the token's own phonetics. The self-supervised model shows none of this (licensed-pool T10 $P(m)$ stays at or below 0.30 beyond $\lambda = 0$), and its $\lambda = 0$ licensed deep cells carry only the known late-layer quirk of the canonical token (a T11–T12 drift that recovers one half-step into the continuum). The modulation is therefore cleanly unidirectional and layer-consistent only in the letter-tuned model, and we attribute it to letter supervision, like the underlying drift itself. We emphasize what the effect is *not*: across the whole 54-item continuum the CTC model's verdicts flip toward the surface reading only at T10–T11 of the licensed frames at the high degrees—never at the phonetic layers, never in the unlicensed pool (NV1 and NV3 return /n/ at 13 of 13 layers at every λ), and the velar drift of NV3/NV4 never surfaces as the trained velar nasal ($P(\text{ŋ}) \leq 0.18$ for NV3 at every layer and degree). Elsewhere the context moves probabilities, not classifications, and only in layers adjacent to the letter output.

7 The Cross-Reference: Our Results × Pouw et al.

We now perform the croisement the title announces: dimension by dimension, what do the two programs agree on, what do they add to each other, and where do they genuinely differ? Table 3 summarizes the comparison; the paragraphs below justify each row.

Convergences. Three results line up so cleanly that the two programs can be said to have measured the same object. The first is surface dominance: their probes read the assimilated consonant as its surface phone at every layer whenever the model's output failed to compensate, and our verdicts read the surface phone in roughly 80 percent of token-layer decisions—despite completely different material (natural recordings vs. synthesis), different probe families (binary TIMIT-trained logistic regression vs. three-way probes trained on our own synthetic canonicals), and different frame-selection rules. When the waveform says [m], the first dozen layers of wav2vec2 say [m]. The second convergence is the late-layer underlying shift restricted to the CTC model: their compensated cases migrate to the underlying phone in the final layers, their pretrained checkpoint (footnote 5) converges instead of diverging, and our factorial design turns that footnote into a systematic contrast—the same drift, present in the letter-tuned model, absent in the self-supervised one. The third is the refusal of the phone code to implement licensing: their unviable compensation survives only at the output level, and our unlicensed controls never recover the underlying phone at the representation level in any layer of either model.

Complementarities. The programs fill each other's blind spots so precisely that the combination is more informative than either alone. Pouw et al. measure *behavior* (transcriptions) and therefore include everything the decoding stage contributes—lexical override of non-words like *cleam*, bigram statistics, the character prior. Our probes measure *representations* at fixed articulatory ground truth and exclude decoding altogether. The instructive case is the unviable condition: their model still "compensates" about 40 percent of unviable non-word stimuli at the output, while our unlicensed [m] before /t/ never recovers /n/ in the representations—not even probabilistically at T4, not even at λ values where the token is mostly [n]. Read together, the two results localize the unviable compensation in the decoding/lexical stage, not

Table 3: Systematic cross-reference between Pouw et al. (2024) and the present experimental program. C = convergent, X = complementary, D = divergent.

Dimension	Pouw et al. (2024)	Present program	
Model set	wav2vec2-base-960h; base checked in a footnote	base-960h and base as the main factorial design	C+X
Stimuli	natural speech, complete assimilation, 3 conditions $\times$ carriers	VTL synthesis, frame-aligned causal pairs, unlicensed controls, 9-step articulatory continuum	X
Default encoding	probing: no-compensation cases stay surface at all layers	$\approx$80% surface across 156 token-layer decisions	C
Late-layer shift	probing: compensated cases shift surface $\rightarrow$ underlying in final layers; pretrained model converges instead	P1 drifts to /n/ at T12 (p=0.56) in the CTC model; base: a single T11 flip	C
Phonological context	ASR: compensation higher in viable than unviable contexts ($\approx$40% residual)	verdicts: no licensing whatsoever (NV controls: /n/ at 13/13 layers at every λ); continuum: no T4 boundary at all; the deep layers read [m] on labial-C2 clusters (P4, with no /m/ segment)	X
Deep-layer context effect	not separable from decoding at the output level	unidirectional labial-context reading at T10–T11: present at $\lambda = 0$ (pooled $P(m) \approx 0.5$ licensed), indifferent to the recruited lip closure, CTC model only	X
Lexical knowledge	output level: non-word vs. real-word asymmetry; *wept*, *playing* over-rides	excluded by design (probing reads representations, no decoding); P3 shows context-lexical conditioning of representations	X
Semantic context	no effect on the model (unlike humans)	not tested	(gap)
Mechanism	interchange interventions: early MLPs at the context phone; heads L4H3/L5H2 move the cue	tract-level intervention: causal delta localizes the divergence in time; depth amplifies it	X
Gradience	excluded by design (complete assimilation only)	the continuum: no assimilation boundary; a graded coarticulation cleanup (velar pools $\rho(\lambda, P(\text{n}))$ up to $+0.98$ at T4)	X

in the phone representation: the encoder hands the decoder an honest [m], and the decoder, disliking the non-word *cleam*, writes *clean*. The continuum adds a finer instrument still: the phone code's category boundary is *context-free* (identical before /p/ and /t/), while the deep CTC layers' underlying-probability is *context-sensitive* (bidirectionally so), which no endpoint-only design could have revealed. Conversely, their causal interchange interventions give what our design cannot: the *components* (early MLPs, heads L4H3 and L5H2) that move the context cue into the assimilated frames, whereas our causal delta gives what theirs cannot: the *time course* and depth profile of a single controlled articulatory change, showing the divergence confined to the manipulated window and amplified monotonically with depth. Their Whisper comparison and our two-model factorial both point at the training objective as the carrier of "compensation knowledge," from opposite directions.

Divergences and open contrasts. Only one dimension shows a real tension, and it is a tension of interpretation rather than of fact. Pouw et al.'s headline is that wav2vec2 exhibits human-like, context-sensitive compensation; ours is that the context sensitivity lives in layers adjacent to the letter output, is probabilistic rather than categorical, and is absent from the self-supervised checkpoint—which their own footnote 5 corroborates. The disagreement is thus about how much weight the word "compensation" should carry when the effect neither flips classifications nor survives without letter supervision. The honest summary is that the two studies bracket the phenomenon: at the output, with the lexicon in the loop, wav2vec2 behaves as a partial compensator; at the representation level, with the lexicon out of the loop, it behaves as an acoustic veridicalist with a supervised, statistical late correction. One contrast remains genuinely untested: semantic context, which their Experiment 3 showed the model ignores; our material (two-word sentences, no biasing contexts) cannot speak to it.

8 A Two-Stage Account of Assimilation Processing

The synthesis of the two programs is best stated as a two-stage architecture, schematized in Figure 5.

Stage 1: an acoustic, context-free phone code (CNN–T6). The early and middle layers of wav2vec2—both checkpoints alike—encode the consonant as what it acoustically is. On the revised continuum this stage is *boundary-free*: the target-level rotation changes the acoustics so little that the probability of the underlying /n/ never crosses 0.5 at any degree of any frame, and what the phonetic layers track is the coarticulation of the canonical cluster (the CNN's /m/-ish baseline) and its cleanup under the velar rotation (ρ up to $+0.98$). This stage is surface-veridical, phonetically graded, and phonologically blind. Its human analogue is the pre-lexical, signal-borne stage that uses residual acoustic traces

Stage 2: a supervised labial-context reading (T7–T12). In the CTC-tuned model only, the deepest layers install a statistical reading shaped by letter supervision: a nasal in a tautosyllabic cluster before a labial is read toward the labial's letter, [m], regardless of the gesture (the expectation effect at $\lambda = 0$ and the surface reading at $\lambda = 1$ are one and the same effect, present on P2/P4 at the baseline, and the full-selector condition shows it is closure-blind: recruiting P1's lips removes its endpoint sharpening and leaves the P2/P4 reading untouched), and its aggregate result is a licensed-pool *deficit* in underlying verdicts—now on both listeners (-20.5 and -17.9 points at $\lambda = 1$, the two significant licensed–unlicensed contrasts we measure)—the opposite sign to the viable-over-unviable advantage that Pouw et al. measured behaviorally. This stage is context-sensitive, unidirectional, probabilistic, and—on the evidence of the self-supervised checkpoint and of Pouw et al.'s footnote—a product of training on underlying spellings rather than of self-supervised structure learning. Its circuit substrate, on the evidence of the interchange interventions, is already in place by layers 4–5, where attention heads gather the context phone and early MLPs write it into the target frames.

What the account buys. The two-stage account explains, in one stroke, every empirical regularity this document has assembled. It explains why surface dominance is the default (stage 1 does the encoding)

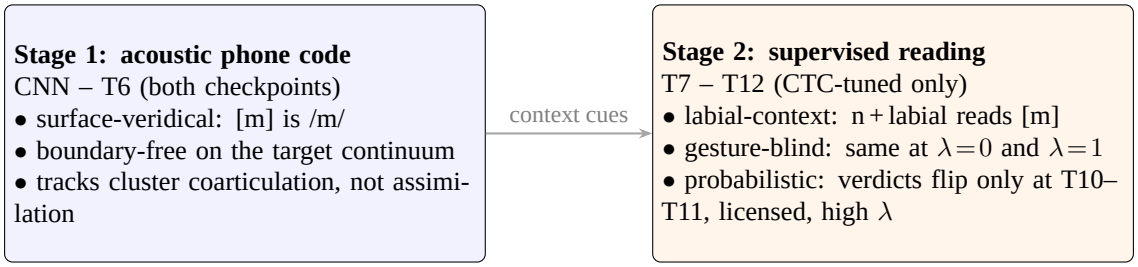


Figure 5: The two-stage account. The division of labor emerges from the cross-reference: an acoustic, context-free phone code, common to both checkpoints, that holds the underlying /n/ across the whole gestural rotation, and a late, letter-supervised, labial-context reading present only in the CTC-tuned model. The context cues that stage 2 exploits are written into the target frames by the circuit that Pouw et al.'s interchange interventions localize: early MLPs and attention heads in layers 4–5.

and why the underlying drift is late, partial, and pair-specific (stage 2 is a statistical correction, not a rule). It explains why unlicensed assimilations are never compensated in the representations (stage 2 moves probabilities, and only where training statistics support them) yet are compensated 40 percent of the time at the output (the decoding stage adds the lexicon). It explains why the phone code has no assimilation boundary on the revised continuum (stage 1 tracks acoustics, and the target rotation barely changes them) while the deep-layer probability is context-sensitive (stage 2 reads the cluster context, not the gesture). And it aligns the machine with the human literature in a specific, testable way: human compensation also has an early, signal-driven component that exploits incomplete-assimilation traces [7] and a later, knowledge-based component that checks phonological viability [9, 14], with the timecourse of the two stages measurable in EEG [14]. The machine's stage 2 differs from the human's in its provenance—letter statistics rather than whatever installs phonological knowledge in humans—and in its indifference to semantics, which Pouw et al.'s Experiment 3 demonstrated and which no amount of probing on two-word sentences could rescue.

Table 4: The experiment kit.

Path	Contents
`README.md`	navigation guide and quick-start
`repro/`	the tooling: the eight scripts of Table 5, plus `requirements.tx`
`work/base/`	the 52-item synthesis base: per item the SAMPA text, the 400 Hz tract-and-glottis sequence, the polar control-plane state, and the 44.1 kHz waveform
`work/results/`	stage-1 probing outputs, per model: LOSO accuracies, classifica-

9 Reproducibility: The Experiment Kit

Every number, verdict, and figure in this document is traceable to a machine-readable artifact, and every artifact is regenerable from the released scripts. The release is organized as an *experiment kit*: one self-contained directory that holds the data of both experimental stages, the tooling that produced them, and the complete source of this document. A reader can therefore move in both directions—from any claim back to the measurements that support it, and forward to a fresh replication on their own machine. The kit is deliberately compact (about 25 MB uncompressed): the only components not shipped are the two wav2vec2 checkpoints and the per-sentence feature caches (a few hundred megabytes), both of which are recreated automatically on first use. This section presents the layout, walks through the pipeline script by script, gives the exact commands for a full re-run, and lists the checks that make a re-run verifiable.

9.1 Layout

Table 4 maps the kit to the concepts of Sections 4 and 6. Two conventions hold throughout. First, every script resolves its inputs and outputs relative to the kit root (the parent directory of `repro/`), so the kit can be unpacked anywhere; the environment variable `COVTL_KIT_ROOT` overrides the root for non-standard layouts. Second, every result file is plain JSON with the layer names (`CNN`, `T1`–`T12`) and the item labels spelled out, so every number in this document can be checked in a text editor, without executing any code. The compiled document itself ships at the kit root as `nasal_place_assimilation_synthesis.pdf`.

9.2 The tooling

Table 5 lists the eight scripts in execution order, with their roles and measured runtimes on a laptop-class CPU. Three design properties deserve emphasis. First, the corpus is *data, not code*: the 52 sentences, their canonical and assimilated variants, and the licensing labels live in one declarative module (`sentence_base.py`) that both the synthesis and the probing scripts import, so the corpus definition cannot drift between stages. Second, the engine modification of Section 4 is a dedicated, idempotent patch script that asserts exactly one match before editing the engine's phoneme table and leaves its provenance comment inside the edited source; the whole change is three lines. Third, the continuum experiment of this edition runs on the reference λ model extended by the kit's own additive module (`repro/morphing_fullsel.py`: under λ the whole CC selector is taken into account): the pinned engine checkout (covtl-pipeline `1d15ed7` + the committed corrections series) with the `covtl-lambda` v1.2.0 patch applied, its 53-test suite green; the 54 syntheses carry the model's four run-time guarantees plus the three full-selector checks, and the probing stage reuses the stage-1 instrument unchanged (LOSO curves identical), so continuum and endpoints are measured alike.

9.3 Rerunning the experiment

A complete replication from a clean machine reduces to the command sequence below. Steps 1–3 rebuild the stage-1 material from scratch; they can be skipped if the shipped `work/base/` is trusted, in which case the replication starts at step 4 and needs about fifteen minutes of compute plus the two checkpoint downloads. Steps 4–6 reproduce the new results of Section 6, figure by figure and number by number.

```
# 0. One-time environment: Python >= 3.10, then
pip install -r repro/requirements.txt   # torch, transformers, sklearn, ...
pip install -e covtl-pipeline           # the vtl-synth engine (VTL backend)
python repro/patch_N.py                 # documented 3-line /N/ closure patch

# 1. Rebuild the 52-item base (or keep the shipped work/base/)
python repro/sentence_base.py
python repro/build_base.py

# 2. wav2vec2 features for the base (downloads both checkpoints once)
python repro/extract_features.py
```

Table 5: The tooling, in execution order. Runtimes are for a laptop-class CPU; the feature stages download the two checkpoints on first use.

Script	Role	Time
`patch_N.py`	engine fix: velar closure for `N` (edit of the covtl-pipeline phoneme table)	s
`sentence_base.py`	declarative corpus definition (52 items, four groups)	s
`build_base.py`	synthesize the base: SAMPA → tract/polar/wav per item	≈5 min
`extract_features.py`	wav2vec2 features: 16 kHz audio → 13 hidden states × 2 models, one `.npz` per sentence (not shipped)	≈30 min
`run_probing.py`	LOSO probes, verdicts, probabilities, cosine statistics, causal delta, time-courses	≈2 min
`figures.py`	stage-1 figures (two of them reappear as Figs. 1 and 2)	s
`make_results.py`	slice the campaign probing output into the kit blocks (continuum + the two production-degree subsets, asserted byte-identical)	s
`note_stats.py`, `pouw075_stats.py`, `pouw0625_stats.py`	the statistics of the anchors, the production degrees, and the nine-degree dose-response	s
`lambda_figures.py`, `note_figure.py`	continuum and note figures (Figs. 3 and 4 among them)	s

```
# 3. Stage-1 probing: verdicts, LOSO accuracies, causal delta
python repro/run_probing.py

# 3b. The reference lambda model (covtl-lambda v1.2.0) on the engine
python /path/to/covtl-lambda/patch/apply_patch.py \
    --pipeline-root /path/to/covtl-pipeline

# 4. The continuum campaign: 6 frames x 9 degrees via morph_targets
python repro/build_campaign.py          # synthesize + guarantee checks
python repro/extract_features.py --campaign
python repro/probe_campaign.py          # -> work/lambda/results.json
python repro/make_results.py            # slice into the kit blocks

# 5. Statistics and figures
python repro/note_stats.py              # anchors + pooled cells
python repro/pouw075_stats.py           # production degree 0.75
python repro/pouw0625_stats.py          # production degree 0.625 + dose-response
python repro/figures.py                 # stage-1 figures
python repro/lambda_figures.py          # continuum figures (figL1, figL2)
```

9.4 Verification anchors

Five checks, all recorded in `work/lambda/synth_meta.json` or recomputed by the scripts themselves, let a reader confirm that a re-run reproduces the reported experiment rather than merely something like it. (1) *Canonicity at the origin*: at $\lambda = 0$ every synthesis is bitwise identical to the engine's canonical output for the cluster spelling (`lam0_bitwise_canon`, asserted per item), so the Control condition is the canonical cluster by construction, not by re-synthesis. (2) *Target equality at the endpoint*: at $\lambda = 1$ the C1 polar target equals the C2 target coordinates (`lam1_theta_equals_c2`), and the glottal block is bitwise constant along every continuum (`glottis_bitwise_constant`). (3) *Structural locality*: outside the C1-incident sub-arcs, every frame is bitwise canonical at every λ

(`outside_c1_arcs_bitwise_canon`)—C2, its transitions, and the vowel context never move; the lips and the velum are bitwise constant along every continuum (readable directly from the shipped tract files). (4) *Probe identity*: the campaign is probed by the unchanged stage-1 instrument (leave-one-sentence-out accuracies identical, nasals 1.00 at T2–T4), so continuum and endpoints are measured alike, and the production-degree blocks are byte-identical slices of the continuum (asserted at $|\Delta P| = 0$). (5) *Provenance*: `synth_meta.json` records the engine tree (commit, corrections series, patch version), the per-frame targets, windows, and guarantee outcomes, and a SHA-256 hash of every tract and wav file. One distance is documented rather than hidden: the dotted citation tier of the earlier kits differs from the cluster continuum by design (maximum $|\Delta P(n)|$ of 0.86 CTC, 0.81 self-supervised, over the twelve frames' canon and assimilated dotted forms)—that number is the coarticulation signature of the tautosyllabic cluster, reported in `endpoint_distance` and its maxima in the stats files.

9.5 Where each number lives

For the reader who wants to audit a specific claim, the following map covers every quantitative statement of Sections 5–8.

- Stage-1 verdicts and probabilities (Figs. 1 and 2, the numbers of Section 5): `work/results/<model>/classification.json`, one record per test token per layer with `pred` and `proba`.
- Probe cross-validated accuracies: `work/results/<model>/probe_cv.json`.
- Continuum probabilities (the boundary-free reading of Section 6, Fig. 3): `work/lambda/results.json` → `<model>` → `cells` → `P1_l0.75` → `T4` → `proba.n`, and its neighbors along both axes.
- Pooled condition cells and contrasts (Fig. 4c and the dose-response): the `pools`, `pooled_continuum`, and `pooled_agreement` blocks of `work/lambda/note_stats.json`; the production-degree conditions in `work/lambda075/pouw075_stats.json` and `work/lambda0625/pouw0625_stats.json`.
- Articulatory ground truth (Fig. 4a) and the guarantees: `work/lambda/synth_meta.json` → `frames[*].c1_window_frames`, `checks`, `files` (SHA-256), plus the shipped `tracts/` themselves.

Model provenance, once and for all:

- *Encoders*: the public hub checkpoints `facebook/wav2vec2-base-960h` (CTC-tuned) and `facebook/wav2vec2-base` (self-supervised).
- *Synthesis*: the covtl-pipeline engine (commit `1d15ed7` + the committed corrections series) with the `covtl-lambda` v1.2.0 reference patch plus the kit's full-CC-selector module, driving VocalTractLab, audio at 44.1 kHz; the campaign guarantees (four run-time + three full-selector) are recorded per item in `work/lambda/synth_meta.json`.
- *Analysis*: resampling to 16 kHz with per-utterance normalization; token vectors are window means over the hidden states whose frame centers fall in $[0.58, 0.68]$ s.
- *Probes*: multinomial logistic regressions ($C{=}1$, up to 5000 iterations) on the stage-1 training base—the same instrument for the endpoints, the continuum, and the production degrees (LOSO curves identical across all three).

10 Glossary

Assimilation (place assimilation) A phonological process by which a segment takes over a property of a neighbor. In English place assimilation, a word-final coronal (/n/, /t/, /d/) takes the place of articulation of the following consonant: /n/ $\rightarrow$ [m] before labials, /n/ $\rightarrow$ [ŋ] before velars.

CTC (connectionist temporal classification) A training objective that maps frame-level encoder states to an unsegmented character sequence. Wav2vec2-base-960h owes its letter supervision to a CTC head; the pairing of assimilated acoustics with underlying spellings is what we call the letter-supervision effect.

Causal delta The training-free quantity $\delta^{(\ell)}(t) = 1 - \cos\big(h^{(\ell)}_{\text{canon}}(t), h^{(\ell)}_{\text{assim}}(t)\big)$: how much a single controlled articulatory change moves the representation, layer by layer and frame by frame.

Compensation Recovering the underlying form of an assimilated segment during perception. Human listeners compensate for licensed assimilation largely unconsciously; the question of this document is whether and where wav2vec2 does.

Frame alignment The property that the two variants of a pair share one frame grid (same boundaries, same frame count), so that representation t of one variant and representation t of the other describe the same instant of the same utterance minus one gesture.

Layer names `CNN` denotes the convolutional feature extractor's output; `T1`–`T12` the twelve transformer blocks, numbered from the input. Both models have thirteen hidden states in total.

Letter supervision The training signal of the CTC-tuned checkpoint: frame states are pushed toward the written characters of the training text, whose spelling preserves underlying forms across assimilation.

Licensing (viable / unviable) Whether the phonological context triggers the assimilation. *clea[m] pan* is licensed (assimilation before /p/ is grammatical); *clea[m] tan* is not (/t/ is alveolar, like the underlying /n/). Pouw et al. use viable/unviable; we use licensed/unlicensed for the same contrast.

Linear probe A shallow classifier (here, multinomial logistic regression) trained on frozen representations to read a property— phone identity—out of them. High probe accuracy means the property is linearly present, not that the model computes it.

Morph strength (λ) The interpolation parameter of the articulatory continuum: $\lambda = 0$ is the canonical [n], $\lambda = 1$ the fully assimilated [m], and intermediate values are convex combinations of the two tract sequences.

Polar control plane The engine's two-dimensional articulatory summary: each phoneme is a target point $\rho e^{i\theta}$ that projects onto the 19-parameter tract state, making phoneme identity a single-point edit.

Probe verdict The probe's argmax class for a token at a layer: the phone identity the layer's representation most strongly encodes.

SAMPA A machine-readable ASCII transcription standard (`klin.pan` for *clean pan*); the engine consumes SAMPA strings, and the velar nasal is written `N`.

Surface dominance The empirical pattern that assimilated tokens are encoded as the phone they have acoustically become (≈80 percent of token-layer decisions in our base).

Tract parameters The 19 vocal-tract shape variables of VocalTractLab (tongue, lips, jaw, velum, larynx), sampled at 400 Hz; together with 11 glottis parameters they form the 30-column state that the morph interpolates.

Underlying / surface form The stored, contrast-bearing representation of a morpheme (/n/ in *clean*) versus the pronounced realization in context ([m] in *clea[m] pan*).

VTL (VocalTractLab) A physiological articulatory synthesizer: a geometric vocal-tract model and a coarticulatory control scheme, producing audio and mid-sagittal contours from tract trajectories.

wav2vec2-base vs. base-960h The same self-supervised encoder architecture: *base* is the raw checkpoint, and *base-960h* is the same model further CTC-tuned on 960 hours of labeled LibriSpeech. Their contrast isolates letter supervision.

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